

VOZHEGA, Russian Federation

WMO: 229540

Lat: 60.479N

Lon: 40.188E

Elev: 201

StdP: 98.93

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-25.3	-31.3	0.2	-28.5	-27.7	0.3	-25.1	8.0	-5.4	7.2	-5.4	2.0	30	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	27.7	19.3	25.7	18.3	23.8	17.4	20.7	25.8	19.4	24.0	18.3	22.7	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.1	17.6	12.9	22.0	16.5	12.0	20.7	60.5	25.8	56.2	24.1	52.5	22.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.4	5.8	-30.8	30.0	4.1	1.9	-33.7	31.4	-36.1	32.5	-38.5	33.6	-41.5	35.0		
(5)			-30.9	22.2	4.0	1.9	-33.8	23.6	-36.1	24.7	-38.3	25.8	-41.2	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-11.1	-9.7	-4.4	3.0	9.9	14.4	17.5	14.8	9.5	2.8	-4.0	-7.8
	DBStd	11.19	7.91	7.10	4.86	4.79	5.16	4.39	3.86	3.78	3.78	4.28	6.01	7.03
	HDD10.0	3203	654	552	447	217	67	9	1	4	54	225	422	551
	HDD18.3	5672	913	786	705	461	264	133	62	121	266	481	672	809
	CDD10.0	640	0	0	0	6	64	141	234	153	38	3	0	0
	CDD18.3	66	0	0	0	0	3	15	37	12	0	0	0	0
	CDH23.3	519	0	0	0	1	29	128	277	84	1	0	0	0
	CDH26.7	110	0	0	0	0	2	27	61	20	0	0	0	0
(14) Wind	WSAvg	2.9	2.9	3.1	3.2	2.9	3.0	2.7	2.4	2.5	2.6	3.0	3.0	3.2
(15) Precipitation	PrecAvg	696	48	36	39	38	58	77	73	76	62	66	64	55
	PrecMax	884	91	65	98	98	139	138	161	150	109	105	123	101
	PrecMin	537	18	9	6	12	14	20	11	25	25	15	16	26
	PrecStd	90	16	15	19	20	28	32	33	34	23	23	21	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.4	8.8	20.7	26.6	29.6	31.2	28.9	22.2	15.9	7.6	4.9
		MCWB	0.7	1.3	3.6	12.4	15.8	19.5	21.7	20.4	16.0	11.7	6.9	4.5
	2%	DB	0.6	1.1	5.5	16.2	23.4	26.8	28.2	25.4	19.3	12.0	5.4	2.5
		MCWB	0.2	0.5	1.8	9.4	15.1	18.9	19.9	18.4	14.9	10.1	4.7	2.0
	5%	DB	0.1	0.2	3.4	12.8	20.9	24.4	26.4	23.1	16.9	10.1	4.0	1.0
		MCWB	-0.2	-0.3	1.0	7.3	13.1	17.0	19.2	17.5	13.4	8.8	3.6	0.5
	10%	DB	-0.9	-0.7	1.8	10.2	18.5	22.2	24.4	21.2	15.1	8.7	2.6	0.1
		MCWB	-1.2	-1.4	0.1	5.6	11.7	15.8	18.5	16.5	12.6	7.6	2.2	-0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.7	4.5	12.4	17.9	22.1	22.8	21.7	17.0	12.6	7.1	4.5
		MCDB	1.1	2.1	7.6	20.0	23.8	27.4	28.2	28.0	20.7	14.7	7.4	4.8
	2%	WB	0.2	0.6	2.6	9.7	16.0	19.5	21.1	19.4	15.3	10.5	4.9	2.1
		MCDB	0.4	1.0	4.5	14.8	21.3	24.5	26.6	23.7	18.1	11.6	5.3	2.4
	5%	WB	-0.2	-0.3	1.3	8.0	14.2	18.0	20.0	18.1	14.0	9.1	3.5	0.7
		MCDB	0.0	0.2	2.7	11.7	19.2	23.0	24.7	22.2	16.1	10.0	3.9	0.9
	10%	WB	-1.2	-1.3	0.4	6.1	12.7	16.5	19.0	16.9	12.8	7.6	2.2	-0.2
		MCDB	-0.9	-0.8	1.6	9.7	17.2	20.9	23.3	20.4	14.7	8.6	2.6	0.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.0	7.1	8.4	9.5	9.0	9.0	8.3	6.6	4.1	3.8	4.4
		MCDBR	4.7	4.6	7.9	12.0	12.5	11.9	11.5	11.0	9.7	6.5	4.1	3.4
	5% WB	MCWBR	4.8	4.4	5.3	7.0	6.1	5.5	4.9	5.1	5.5	4.8	4.0	3.4
		MCDWR	4.6	4.7	5.9	11.0	10.9	10.8	10.5	9.8	8.2	6.2	3.9	3.4
(40) Clear-Sky Solar Irradiance	taub		0.254	0.299	0.324	0.365	0.352	0.351	0.383	0.374	0.340	0.321	0.278	0.229
		taud	2.041	2.074	2.105	2.221	2.439	2.462	2.384	2.392	2.474	2.392	2.143	1.987
	Ebn at Noon		477	665	784	826	874	880	836	814	785	665	448	344
		Edh at Noon	44	79	106	115	100	100	106	97	75	59	40	29
(44) All-Sky Solar Radiation	RadAvg		0.30	0.91	2.15	3.70	5.02	5.30	5.25	3.85	2.20	0.91	0.28	0.15
	RadStd		0.05	0.21	0.30	0.39	0.32	0.40	0.51	0.29	0.23	0.13	0.05	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-199	-245	N/A	N/A

Nomenclature: See separate page